

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 3 – Tree Traversal-Based Questions

Course Code:	CSA03	Course Title:	Data Structures
CO Assessed:	CO3 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 3- Tree Traversal Based Questions
Weightage:	10% of CIA	Total Marks:	50 (5 Questions × 10 Mark)
CO5 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity (BL3, BL4)		
Student Name:	T.VAISHNAVI	Reg. No.:	192421370
Section / Batch:	B-TECH(IT)	Date:	10/08/2026

Code of Conduct

I **T.VAISHNAVI(192421370)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Instructions

- This test contains 5 Traversal Questions Total: 50 Marks.
- Evaluation of Answer Quality - Understanding of Tree Structure, Traversal Execution, Procedure, Accuracy & Logical Reasoning and Complexity Analysis
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Annexure A – Traversal Based Questions

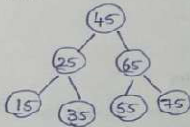
C03-AT3

T. Vaishnavi (192421370)

Data Structure

- ① Construct the Binary Search Tree 45, 25, 15, 35, 65, 55, 75 of Pre-order Inorder, Post order, Level order?

A) Binary Search Tree



(i) Preorder [Root $\rightarrow$ Left $\rightarrow$ Right]

Rule:- First visit Root, then Left, then Right.

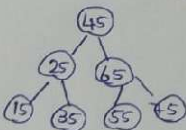


Traversals:- 45 $\rightarrow$ 25 $\rightarrow$ 15 $\rightarrow$ 35 $\rightarrow$ 65 $\rightarrow$ 55 $\rightarrow$ 75

Justification:- Start from root 45, then visit the left subtree 25 $\rightarrow$ 15 $\rightarrow$ 35. Next visit the right subtree 65 $\rightarrow$ 55 $\rightarrow$ 75.

(ii) Inorder [Left $\rightarrow$ Root $\rightarrow$ Right]

Rule:- First visit Left then Root, then Right

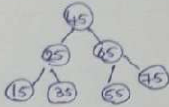


Traversals:- 15 $\rightarrow$ 25 $\rightarrow$ 35 $\rightarrow$ 45 $\rightarrow$ 55 $\rightarrow$ 65 $\rightarrow$ 75

Justification:- Start from the leftmost node 15, then visit 25 $\rightarrow$ 35, followed by root 45. Then traverse the right subtree: 55 $\rightarrow$ 65 $\rightarrow$ 75, following Left $\rightarrow$ Root $\rightarrow$ Right.

(iii) Post orders [Left $\rightarrow$ Right $\rightarrow$ Root]

Rule:- First visit left, then Right, and finally Root

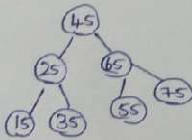


Traversals :- $15 \rightarrow 35 \rightarrow 25 \rightarrow 55 \rightarrow 75 \rightarrow 65 \rightarrow 45$

Justification:- First traverse the left subtree: 15 → 35 → 25. Then traverse the right subtree: 55 → 75 → 65. Finally visit the root node 45.

(iv) Level-order [Level by Level]

Rule:- Visit nodes from top to bottom, left to right



Traversals:- $45 \rightarrow 25 \rightarrow 65 \rightarrow 15 \rightarrow 35 \rightarrow 55 \rightarrow 75$

Traversal:- $45 \rightarrow 25 \rightarrow 65 \rightarrow 15 \rightarrow 35 \rightarrow 55 \rightarrow 75$
Justification:- Visit nodes level by level from top, to bottom and left to right. So, $45 \rightarrow 25 \rightarrow 65 \rightarrow 15 \rightarrow 35 \rightarrow 55 \rightarrow 75$.

② Construct a Binary Search Tree.

Incodez - DBA EFCH

Pre-order - A B D E C F G

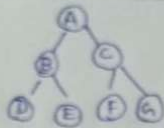
(A)

The first node in preorder is A, so A is the root

Inorder: D B E | A | F C G

Left Sub tree

Right subtree



Post order (Left $\rightarrow$ Right $\rightarrow$ Root)

$D \rightarrow E \rightarrow B \rightarrow F \rightarrow G \rightarrow C \rightarrow A$

Level order Traversal:-

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

③ Construct a Binary Search Tree

40 20 60 10 30 50 70 25 35

A) Binary Search Tree:-



(i) Inorder (Left $\rightarrow$ Root $\rightarrow$ Right)

$10 \rightarrow 20 \rightarrow 25 \rightarrow 30 \rightarrow 35 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 70$

Justification:- Visit the left subtree first, then root, and finally the right subtree.

(ii) Preorder (Root $\rightarrow$ Left $\rightarrow$ Right)

$40 \rightarrow 20 \rightarrow 10 \rightarrow 30 \rightarrow 25 \rightarrow 35 \rightarrow 60 \rightarrow 50 \rightarrow 70$

Justification:- Visit the root first, then traverse the left subtree followed by the right subtree.

(iii) Postorder (Left $\rightarrow$ Right $\rightarrow$ Root)

10 $\rightarrow$ 25 $\rightarrow$ 35 $\rightarrow$ 30 $\rightarrow$ 20 $\rightarrow$ 50 $\rightarrow$ 70 $\rightarrow$ 60 $\rightarrow$ 40

Justification :- Visit the left and right subtrees first, then visit the root.

(iv) Level order (Level by Level)

40 $\rightarrow$ 20 $\rightarrow$ 60 $\rightarrow$ 10 $\rightarrow$ 30 $\rightarrow$ 50 $\rightarrow$ 70 $\rightarrow$ 25 $\rightarrow$ 35

Justification :- Visit each level from top to bottom and from left to right.

4) Construct a Binary Search Tree

Pre-order = 1, 2, 4, 5, 3, 6, 7

In-order = 4, 2, 5, 1, 6, 3, 7

A) In Pre-order, the first element is the root (Root = 1)

In In-order, left subtree = 4, 2, 5

right subtree = 6, 3, 7

Left subtree

Pre-order for left = 2, 4, 5 (Root = 2)

In-order: 4, 2, 5

left child = 4

right child = 5

Right subtree

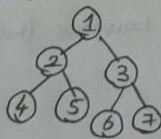
Pre-order for right = 3, 6, 7

Root = 3

In-order: 6, 3, 7

left child = 6

right child = 7



Pre order :-

1 → 2 → 4 → 5 → 3 → 6 → 7

Inorder :-

4 → 2 → 5 → 1 → 6 → 3 → 7

- ⑤ Construct Binary Tree Inorder : H, D, I, B, E, A, F, C, G, Postorder : H, I, D, E, B, F, G, C, A

A) In Postorder the last element is the root

Root = A

Now locate A in inorder : H D I B E A F C G

Therefore, left subtree

Inorder : H, D, I, B, E

Postorder : H, I, D, E, B

Right subtree

In order : F, C, G

Post order : F, G, C

left subtree

last element of Postorder is the root, Root = B

locate B in inorder : H D I B E

Therefore, left subtree of B

In order : H, D, I

Post order : H, I, D

Right subtree of B

Inorder : E

Post order : E

last element of Postorder is the root, Root = D

locate D in inorder : H D I

Therefore, left subtree of D

Inorder : H



Post order: H

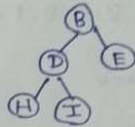
So, H is the left child of D

Right Subtree of D: Inorder: I

Post order: I

So, I is the right child of D

Left subtree becomes



Right subtree

last element of Post order is the root

Root = C

locate C in inorder: F < C < G

Inorder: F, Postorder: F

So, F is the child of C

Right subtree of C

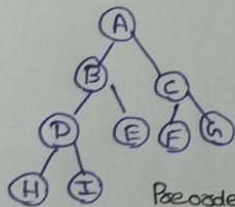
Inorder: G, Postorder: G

So, G is the right child of C

Right subtree becomes



Final Tree:-



Preorder rule :- Root $\rightarrow$ left $\rightarrow$ Right

first root: A

left subtree: B, D, H, I, E

Right subtree: C, F, G

$\therefore$ Therefore, Preorder = A, B, D, H, I, E, C, F, G